

| Question |     |       | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Marks available |        |     |       | Maths | Prac |
|----------|-----|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|--------|-----|-------|-------|------|
|          |     |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | AO1             | AO2    | AO3 | Total |       |      |
| 6        | (a) |       | <b>Vertical:</b> Decreasing (accept deceleration), then increasing (accept acceleration) / <u>changes</u> at $9.81 \text{ m s}^{-2}$ (1)<br><b>Horizontal:</b> Constant (1)<br><b>Reason:</b> Gravity acts vertically <b>or</b> no forces act horizontally (1)                                                                                                                                                                                                                                     | 3               |        |     | 3     |       |      |
|          | (b) | (i)   | $0.15 [\text{m s}^{-1}]$ i.e. $\frac{(1000)}{(110 \times 60)}$ or $0.54 \text{ km/h}$                                                                                                                                                                                                                                                                                                                                                                                                              |                 | 1      |     | 1     |       |      |
|          |     | (ii)  | Correct substitution into $x = ut + \frac{1}{2}at^2$ Ignore sign convention<br>e.g. $1000 = \frac{1}{2} \times a \times (55 \times 60)^2$ (1)<br>At least one mathematical step shown leading to<br>$a = 0.00018 [\text{m s}^{-2}]$ e.g. $a = \frac{2000}{1.09 \times 10^7}$ (1)<br><b>Alternative:</b><br>$u_{\text{vertical}}$ calculated from $x = \frac{1}{2}(u + v)t$ i.e. $u = 0.606 \text{ m s}^{-1}$ (1)<br>Substitution into: $a = (v - u)/t$ to show $a = 0.00018 [\text{m s}^{-2}]$ (1) | 1               | 1      |     | 2     | 2     |      |
|          |     | (iii) | Correct substitution into $v = u + at$ <b>or</b> $v^2 = u^2 + 2ax$<br>e.g. $0 = u - 0.00018 (55 \times 60)$ <b>or</b> $0 = u^2 - 2 \times 0.00018 \times 1000$<br>(1) <b>ecf</b> [accept use of $0.0002 \text{ m s}^{-2}$ ]<br>e.g. $u = 0.61 [\text{m s}^{-1}]$ (1)<br>e.g. $\frac{0.61}{0.88} \times 100\%$ seen (1) Accept $67\% - 75\%$<br><b>Alternative for final mark:</b><br>$60\% \text{ of } 0.88 \text{ m s}^{-1} = 0.53 \text{ m s}^{-1}$ therefore: $0.61 > 0.53$                     | 1               | 1<br>1 |     | 3     | 3     |      |

| Question |     |  | Marking details                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                    | Marks available |     |     |       |       |      |
|----------|-----|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |  |                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                    | AO1             | AO2 | AO3 | Total | Maths | Prac |
|          | (c) |  | For                                                                                                                                                                                                                                                 | Against                                                                                                                                                                                                                            |                 |     | 3   | 3     |       |      |
|          |     |  | <ul style="list-style-type: none"><li>• Job creation</li><li>• Cost/year reasonable</li><li>• Generate interest in science</li><li>• New technologies developed e.g. renewable</li><li>• Improve understanding of origin of life on earth</li></ul> | <ul style="list-style-type: none"><li>• Funding could have been used to address earth based issues.</li><li>• Little impact on society</li><li>• Costs outweigh discoveries</li><li>• Risky mission it might have failed</li></ul> |                 |     |     |       |       |      |
|          |     |  | 3 statements given <b>must expand</b> on bullet points in question (3)<br>No mark for agreeing or disagreeing                                                                                                                                       |                                                                                                                                                                                                                                    |                 |     |     |       |       |      |
|          |     |  | Question 6 total                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                    | 5               | 4   | 3   | 12    | 5     | 0    |